gravitational fields

field $\implies$ when two separated objects exert a force on each other

Gravitational **field lines** show the direction of the force acting on a mass in a gravitational field.

Uniformed gravitational field $\implies$ field lines are **parallel** and **equally spaced**.

Roughly speaking, Earth has a radial gravitational field that is uniform.

Gravitational field strength = the force per unit mass that acts on a mass in a gravitational field. See Newton's law of gravitation.

The (real) objects w.r.t gravitational theory are known as **extended bodies**.

When two extended bodies interact gravitationally, they can be treated **as point masses** by assuming that **the whole of the mass of one object** is placed **at its centre of gravity**.

This also assumes that the objects are in a uniform gravitational field or that they are well away from any other mass.

When an object is in a non-uniform gravitational field, the system cannot be treated as a series of point masses.

Equipotential lines must always meet field lines at 90°. **No work** is done when a charge or mass moves along an equipotential line.

The **potential** **gradient** is the **field strength** $\iff \frac{\text{change in potential}}{\text{change in distance}} = -\frac{\Delta V_g}{\Delta r}$.